

YANG WANG

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6895 Shooting Star Dr ◇ Whitestown, IN 46075

EDUCATION

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| Purdue University, West Lafayette
Ph.D. in Electrical Engineering
<i>Thesis: Design and Implementations of Open-Source Ag IoT Devices for Farm Machinery Data Acquisition and Integrated Analytics</i> | Aug. 2015 – Jun. 2021 |
| Purdue University, West Lafayette
B.S. in Electrical Engineering | Aug. 2010 – Dec. 2014 |

EXPERIENCE

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| The Climate Corporation
<i>Senior Data Scientist</i> | Nov. 2022 – Present
<i>Remote</i> |
| <ul style="list-style-type: none">· Collaborated cross functionally to build a scalable geospatial data processor enabling efficient spatial joins and aggregation of cross-activity operational datasets using DuckDB and Apache Hamilton.· Drove architectural decisions and implemented core data models and backend processing components for a Yield Correction API, including development of a Python-based AWS SQS consumer and containerized deployment to AWS ECS with infrastructure provisioned via Terraform.· Led development of an AWS ECS service to convert shapefiles into an internal geospatial format, increasing conversion success rate from 50% to 95% and eliminating ongoing 3rd-party service costs.· Architected and implemented a spatiotemporal processing pipeline using connected components algorithms to aggregate vector data, achieving up to 10x data size reduction with minimal fidelity loss. | |
| John Deere Intelligent Solutions Group
<i>Senior Navigation Engineer</i> | Jul. 2021 – Oct. 2022
<i>Torrance, CA</i> |
| <ul style="list-style-type: none">· Refactored and enhanced GNSS receiver (StarFire) firmware in C++, improving reliability and maintainability of embedded navigation systems. | |
| Purdue University
<i>Graduate Research / Teaching Assistant</i> | Aug. 2015 – Jun. 2021
<i>West Lafayette, IN</i> |
| <ul style="list-style-type: none">· Architected and developed open-source agricultural telematics units (ISOBlue 2) running Yocto Linux for farm machinery data acquisition. Deployed 10+ units across multiple U.S. states and internationally, coordinating directly with farmers and operators.· Implemented Kalman filtering and spatiotemporal clustering algorithms for GNSS track analysis and harvest productivity modeling. | |
| Spensa Technologies Inc. (Acquired by DTN)
<i>Embedded Systems Engineer</i> | Jan. 2015 – May. 2015
<i>West Lafayette, IN</i> |
| <ul style="list-style-type: none">· Collaborated across hardware and software teams to deliver production-ready embedded systems. | |

CORE COMPETENCIES

Programming Languages	Python, C, C++, MATLAB, Shell scripting, $\text{\LaTeX}$, Java, Javascript
Cloud & Orchestration	AWS, Docker, Apache Hamilton
Embedded & Edge Systems	Yocto Linux, Android, ISOBUS VT, IoT
Network & Messaging Protocols	CAN, J1939, ISOBUS, MQTT, Apache Kafka
Databases & Observability	MySQL, TimescaleDB, DuckDB, Datadog, OpenSearch, Splunk
Version Control Systems	Git, Team Foundation Server
Languages	English and Mandarin (both fluent), French (limited proficiency)